

MyDatabasesTheory

Database Theory – Practical Task

Question 1 — Data vs Information

Explain the difference between data and information

Data refers to raw facts and figures that have not yet been processed or organised. Data can include numbers, names, dates, measurements or other individual values. On its own, data may not provide much meaning.

Information is data that has been processed, organised or interpreted so that it becomes meaningful and useful. Information can help people understand situations, identify patterns and make decisions.

For example, a list of individual sales amounts is data. When those sales amounts are processed to calculate the total sales for a month, the result becomes information because it provides a useful understanding of the company's sales performance.

Question 2 — What is metadata?

What is metadata?

Metadata is information that describes or provides details about other data. It is often referred to as "**data about data**." Metadata can describe the characteristics, structure, or properties of data.

For example, information about a photograph such as its file name, file size, date created, file type and dimensions is metadata. In a database, metadata can include information about table names, column names, data types and relationships between tables. Metadata helps users and database systems understand how data is organised and what the data represents.

Question 3 — What is a DBMS?

What is a DBMS, and what are its advantages?

A **Database Management System (DBMS)** is software used to create, store, organise, manage and retrieve data in a database. It provides users and applications with tools to work with data efficiently and securely.

There are several advantages to using a DBMS. It helps organise large amounts of data, improves data security by controlling access to information, and helps maintain data accuracy and consistency. A DBMS also allows data to be shared between multiple users or applications and can provide backup and recovery features to help protect data from being lost.

Examples of DBMSs include MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database and SQLite.

Question 4 — Operational vs Analytical databases

Compare and contrast operational databases with analytical databases, and provide an example of each.

Operational and analytical databases are both used to store and work with data, but they are designed for different purposes.

Operational databases are designed to support the day-to-day activities of an organisation. They handle frequent transactions such as adding, updating and deleting records. For example, a restaurant could use an operational database to store customer orders, payments, menu items and stock levels.

Analytical databases are designed to support data analysis and decision-making. They are mainly used to examine large amounts of data, identify trends and produce useful insights. For example, a restaurant could use an analytical database to analyse its sales history and determine which meals are the most popular.

The main difference is that an operational database focuses on **running daily business operations**, while an analytical database focuses on **analysing data to support decision-making**.

Question 5 — Types of data and NoSQL Databases

Explain the types of data and use cases where NoSQL databases are most effective.

Data can be classified into three main types: structured, semi-structured and unstructured data.

Structured data has a clearly defined structure and is usually organised into rows and columns. Examples include customer records, employee information and financial transactions.

Semi-structured data does not follow the traditional table structure of a relational database, but it still contains some organisation or structure. Examples include JSON and XML files.

Unstructured data does not have a predefined structure. Examples include images, videos, audio files, documents and other forms of multimedia.

NoSQL databases are most effective when working with large amounts of data that may have a flexible or changing structure. They are useful for applications that need to handle rapidly growing data and scale across multiple servers. For example, social media applications can use NoSQL databases to manage large amounts of user profiles, posts, comments and other types of content.

Question 6 — DBMS that doesn't require server configuration

Which DBMS does not require server configuration, and what are its advantages?

SQLite is a database management system that does not require server configuration because it is a serverless database system. Unlike database systems that require a separate server to be installed and managed, SQLite stores the database directly in a file.

SQLite has several advantages. It is lightweight, easy to set up and requires very little configuration. It is also portable because the database can be stored in a single file and moved between systems. SQLite is suitable for small applications, development and testing, and embedded applications where a separate database server is not necessary.

Question 7 — ACID properties

Explain the ACID properties.

ACID represents four important properties that help ensure database transactions are reliable and maintain the integrity of the data. ACID stands for Atomicity, Consistency, Isolation and Durability.

Atomicity means that a transaction must be completed entirely or not completed at all. If one part of a transaction fails, the entire transaction should be rolled back.

Consistency means that a transaction must leave the database in a valid and consistent state. Database rules and constraints must continue to be maintained after the transaction is completed.

Isolation means that transactions occurring at the same time should not incorrectly interfere with one another. Each transaction should operate independently so that the data remains accurate.

Durability means that once a transaction has been successfully completed, the changes are permanently saved. The data should remain available even if there is a system failure after the transaction has been completed.
